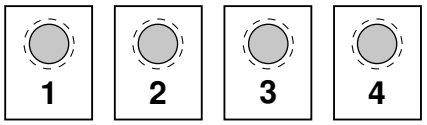
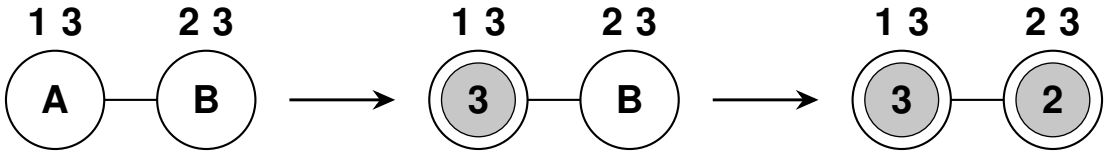
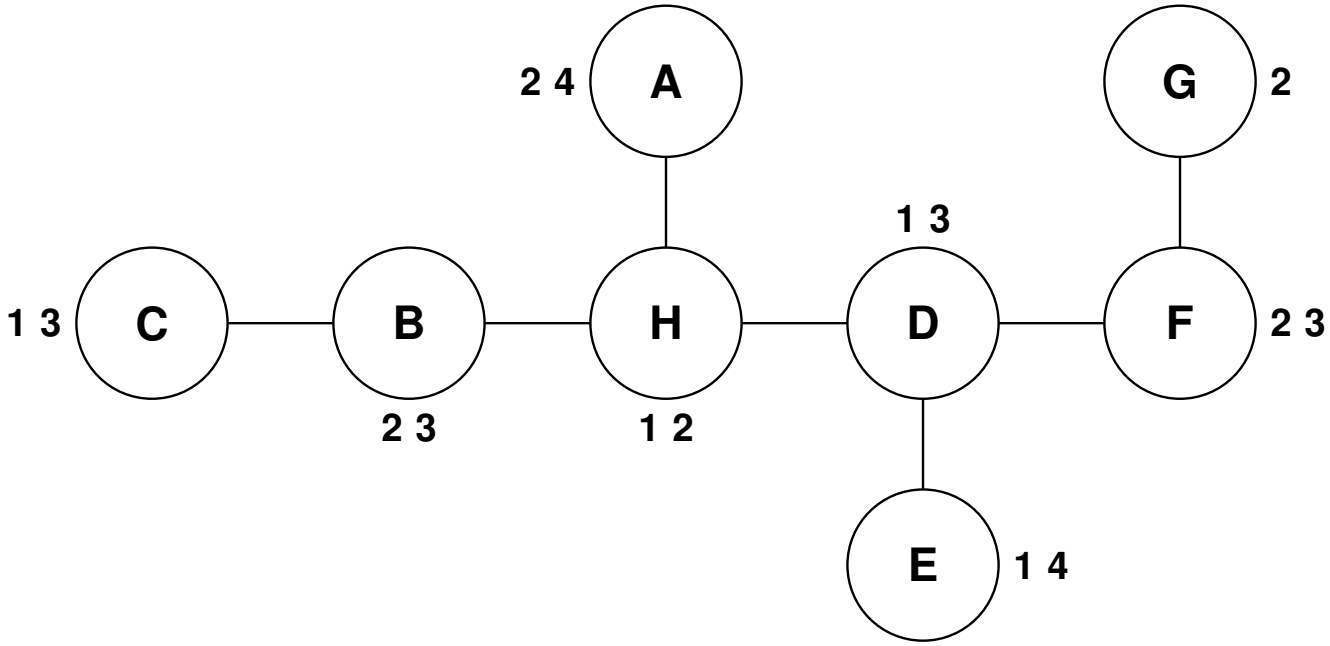
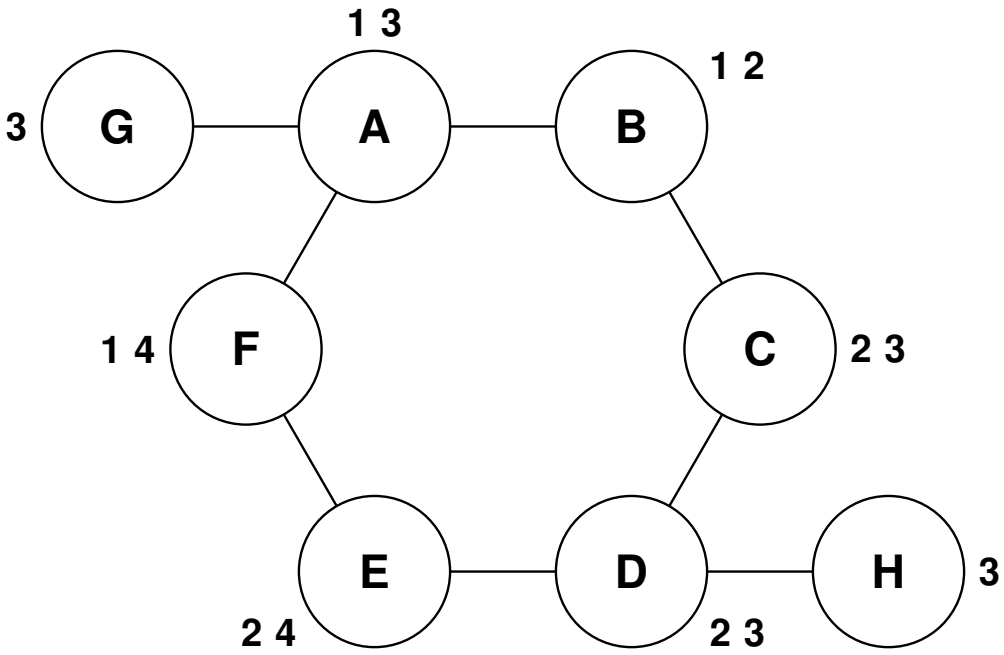
The headers show which colour goes with each number. Put one counter on each circle. Two circles joined by a line must not get the same number. When numbers are printed next to a circle, its counter must be one of them.



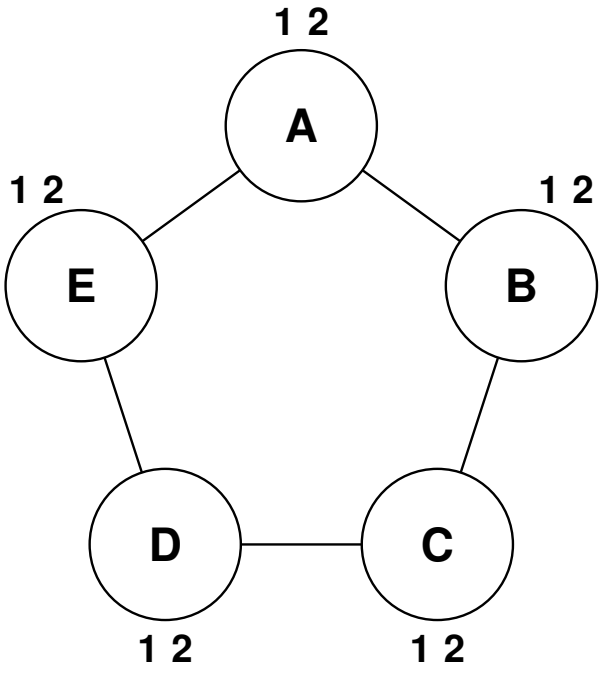



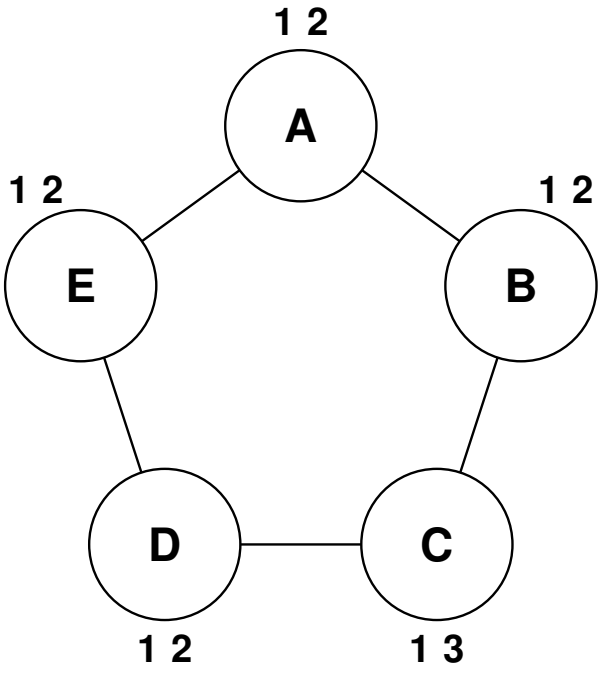

**Problem 1:** Put a counter on every circle, or explain why it can't be done.

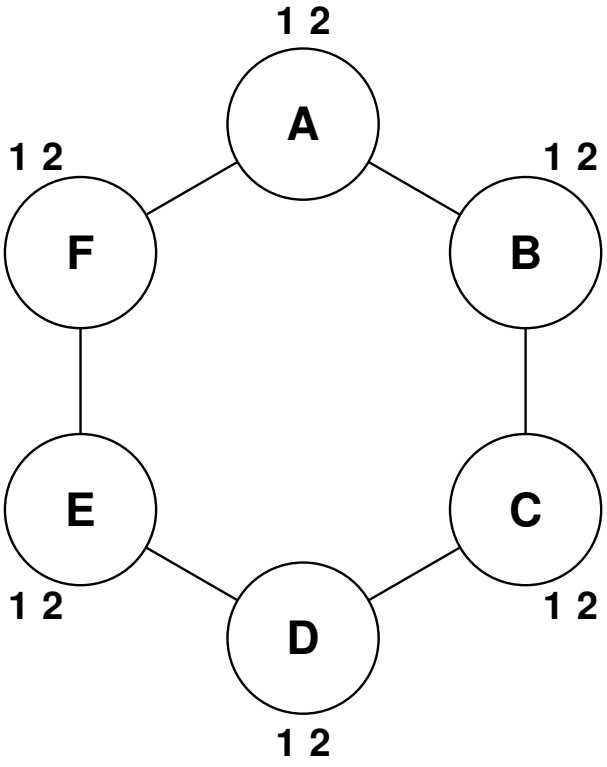


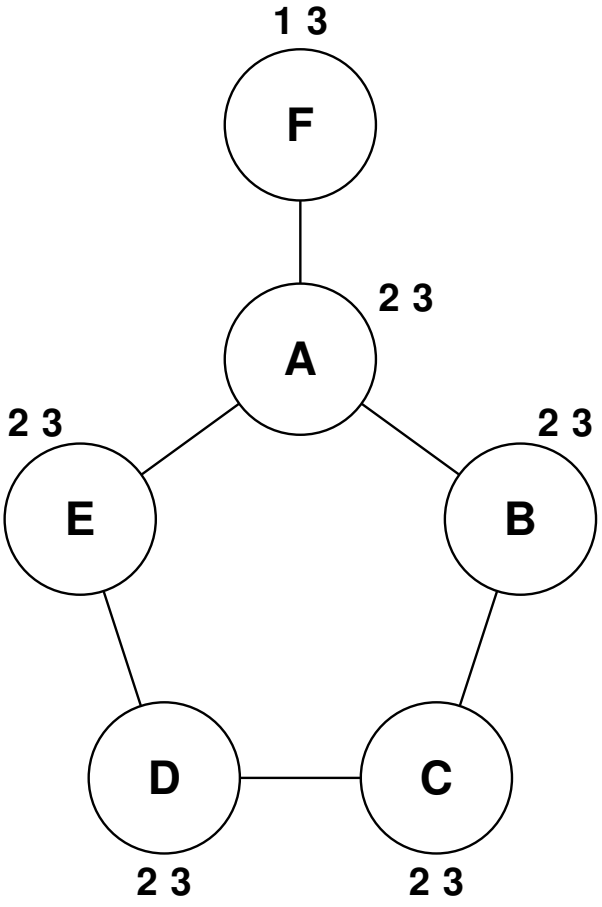



**Problem 2:** Which of these can be done?

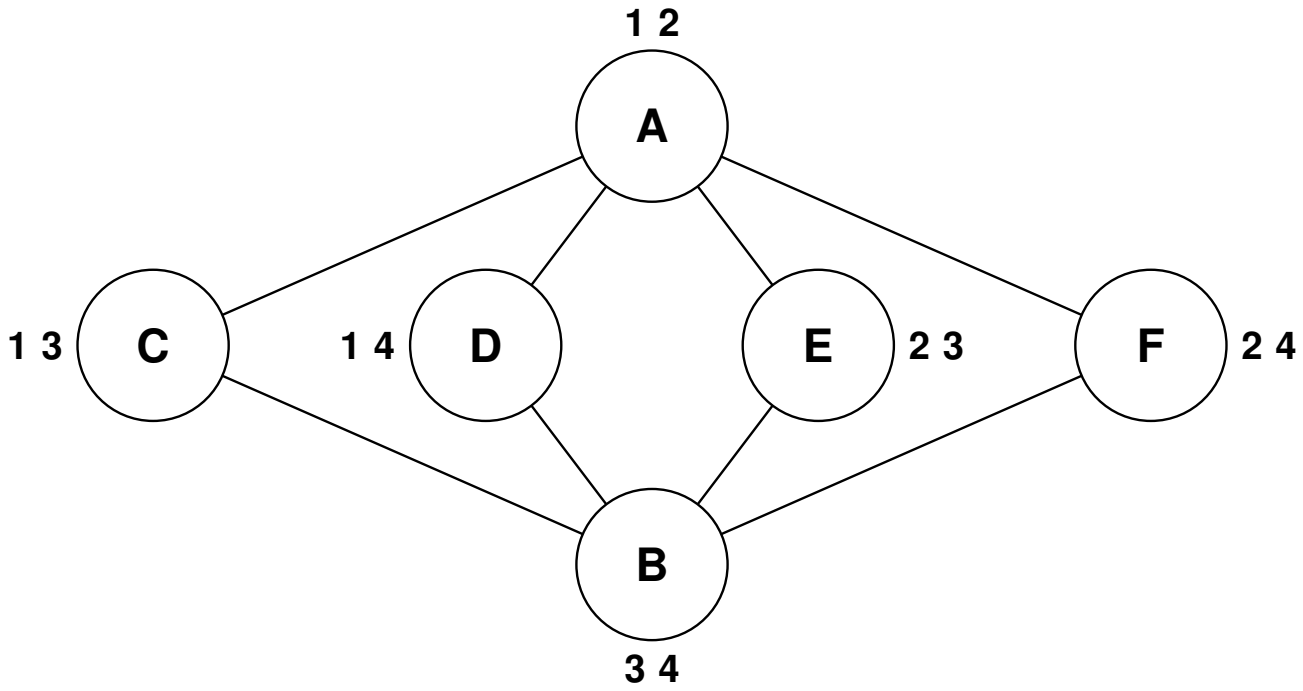


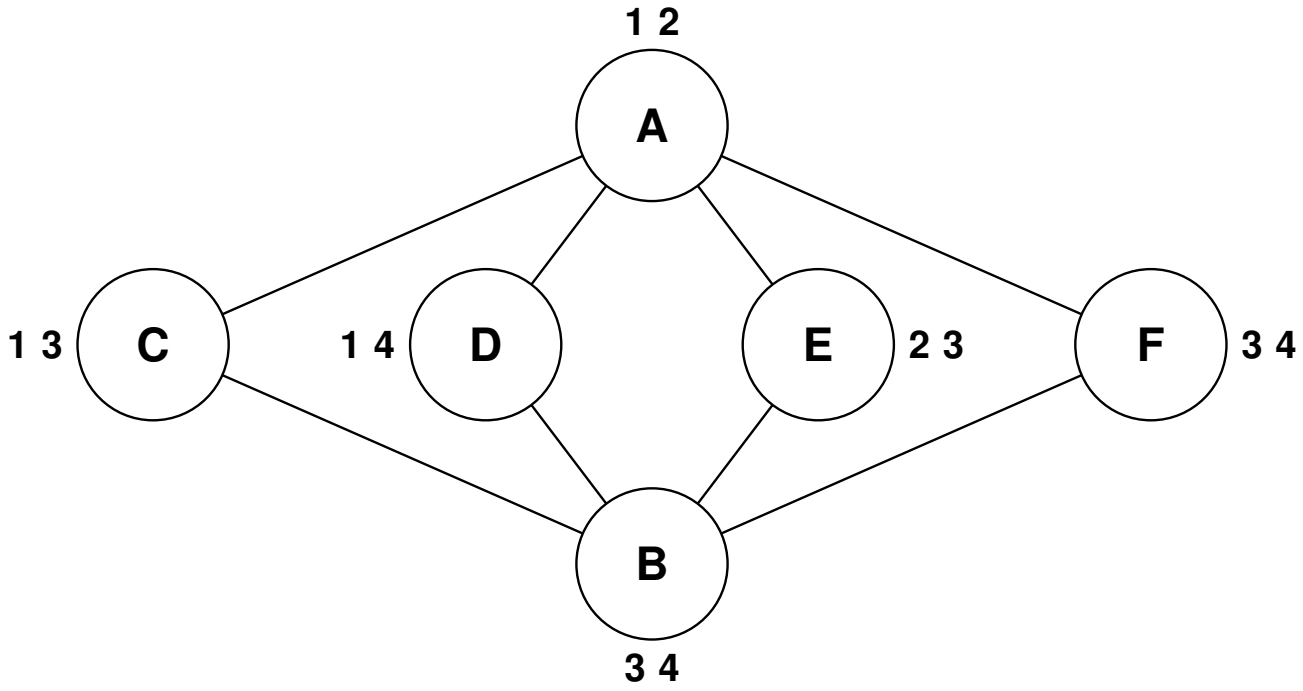



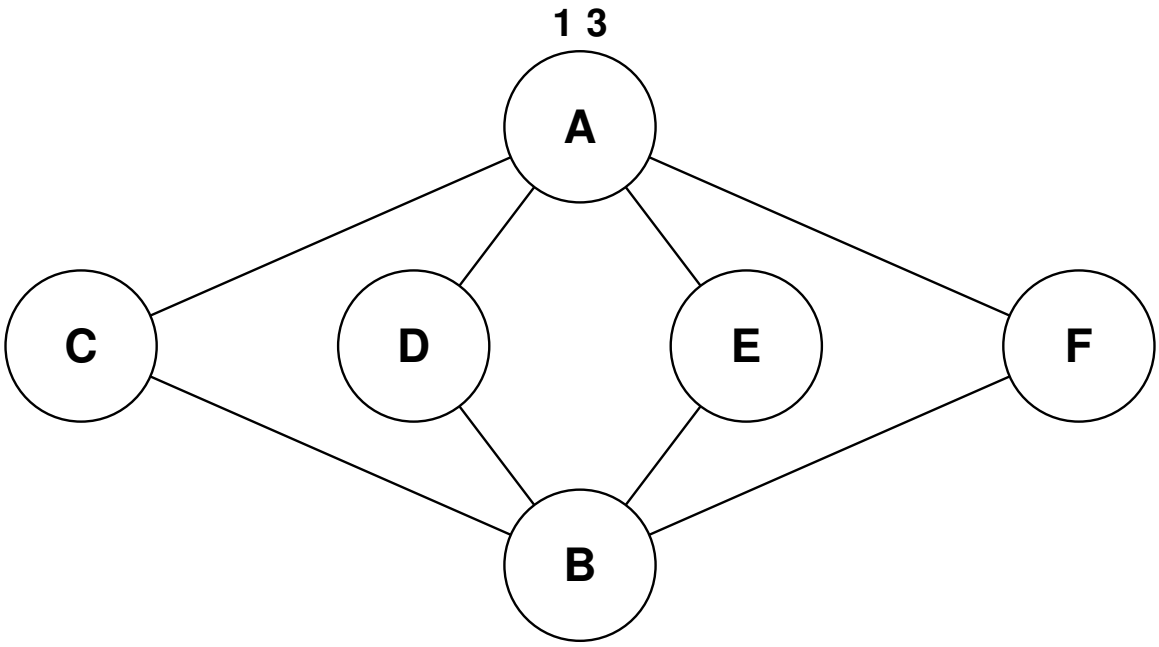



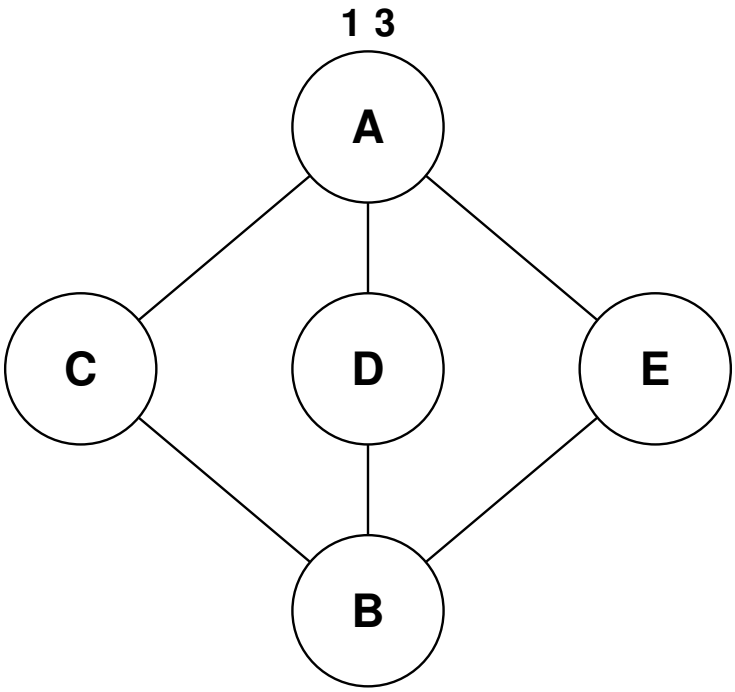
**Problem 3:** Which of these can be done?



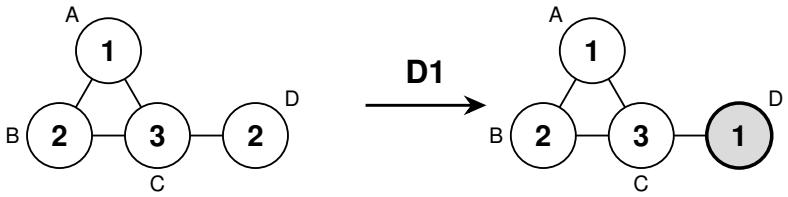



**Problem 4:** For each network, can you write two of the numbers 1, 2, 3, 4 next to every circle without numbers, so that it can't be done?

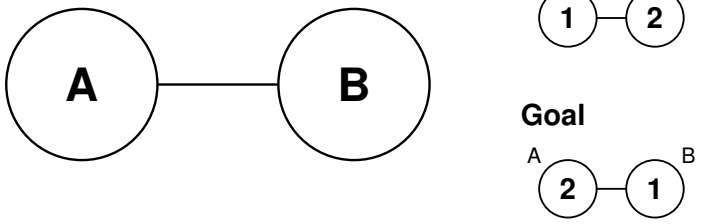


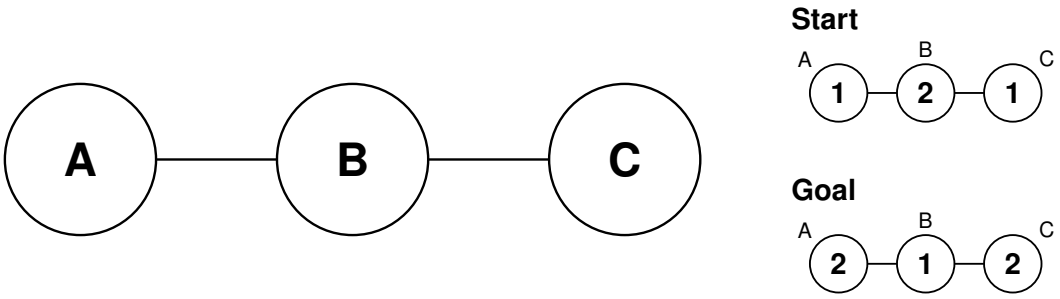
A move takes one counter off a circle and puts a counter with a different number there. After every move, no line may have the same number at both ends. Write a move as the circle's letter and its new number, like D1.




**Problem 5:** Use counters 1, 2 and 3. Set up Start, then change it into Goal in as few moves as you can. Write your moves.

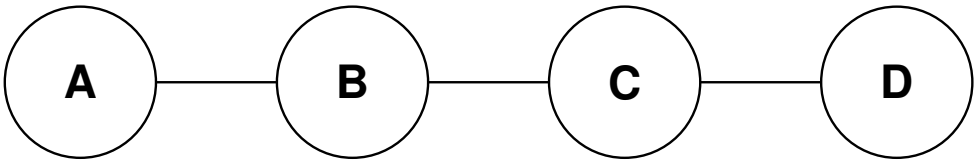



Moves: _____

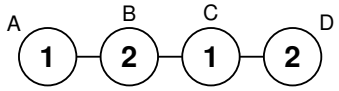



Moves: _____

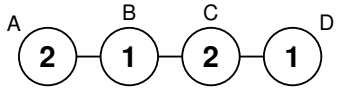
**Problem 6:** Use counters 1, 2 and 3. Change each Start into its Goal in as few moves as you can. Write your moves.



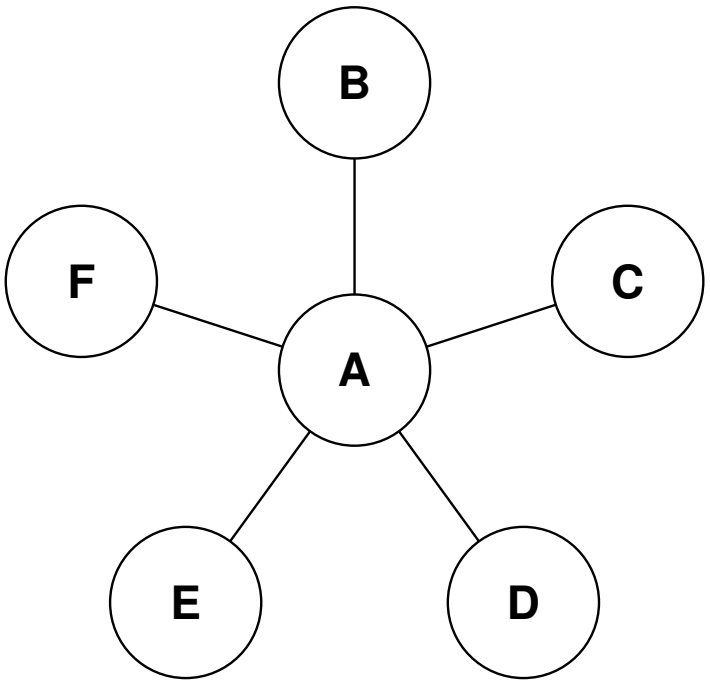

**Start**



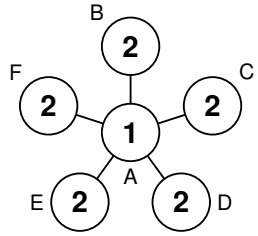

**Goal**



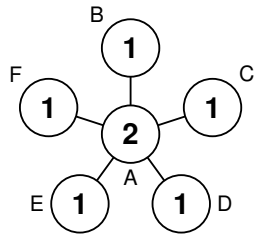

Moves: _____




**Start**



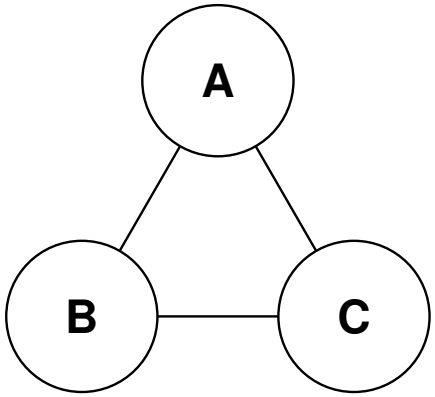

**Goal**



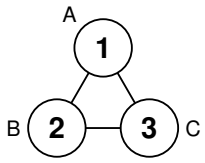

Moves: _____

_____

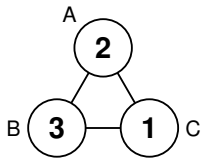
**Problem 7:** Using only the counters shown, can you change Start into Goal? Write the fewest moves you can find, or explain why it can't be done.




**Start**

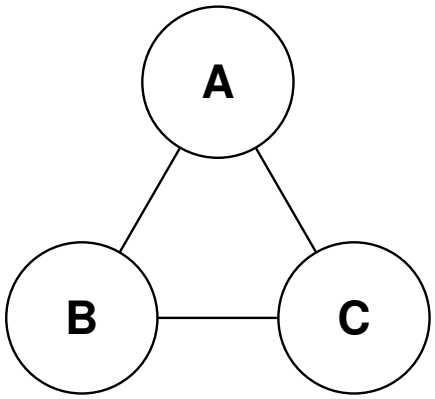



**Goal**

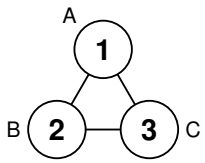



Counters 1 2 3

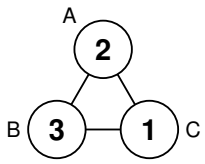
Moves: _____




**Start**



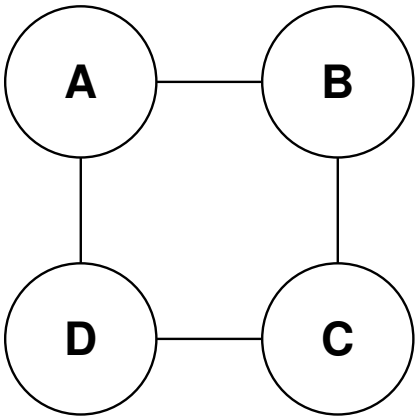

**Goal**

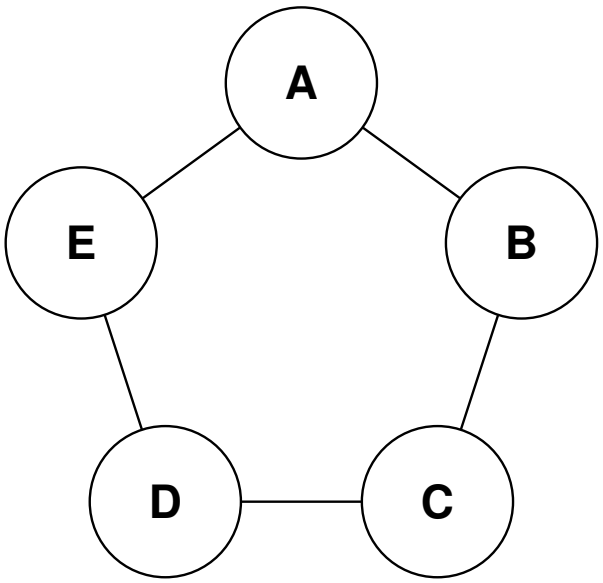


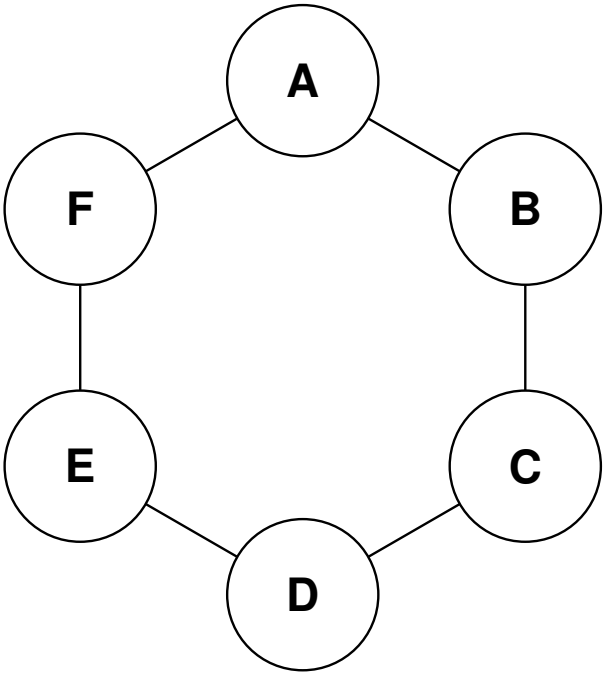

Counters 1 2 3 4

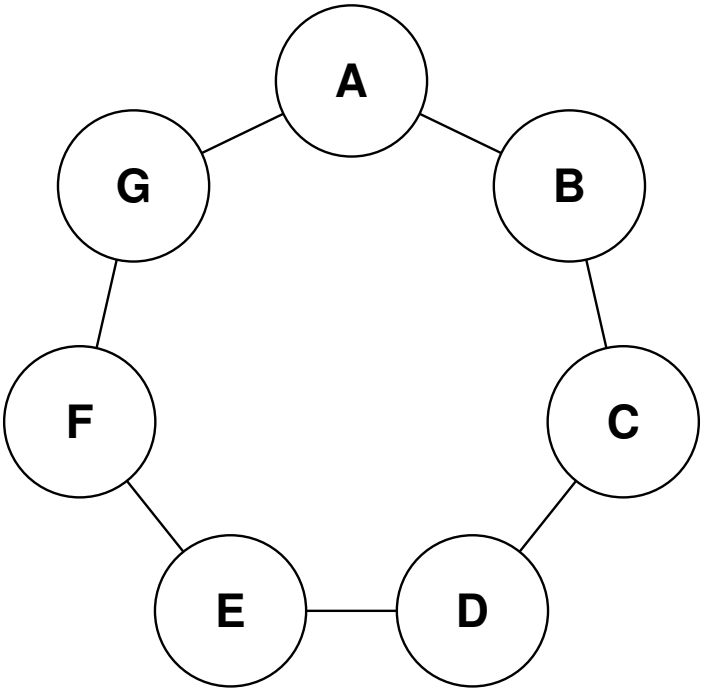
Moves: _____

**Problem 8:** Use counters 1, 2 and 3. On which of these rings can you place counters, with different numbers at the two ends of every line, so that no counter can move?

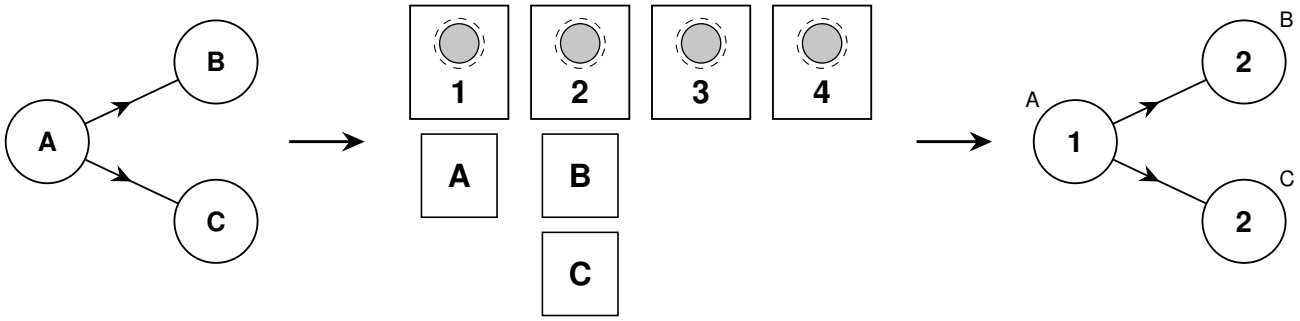


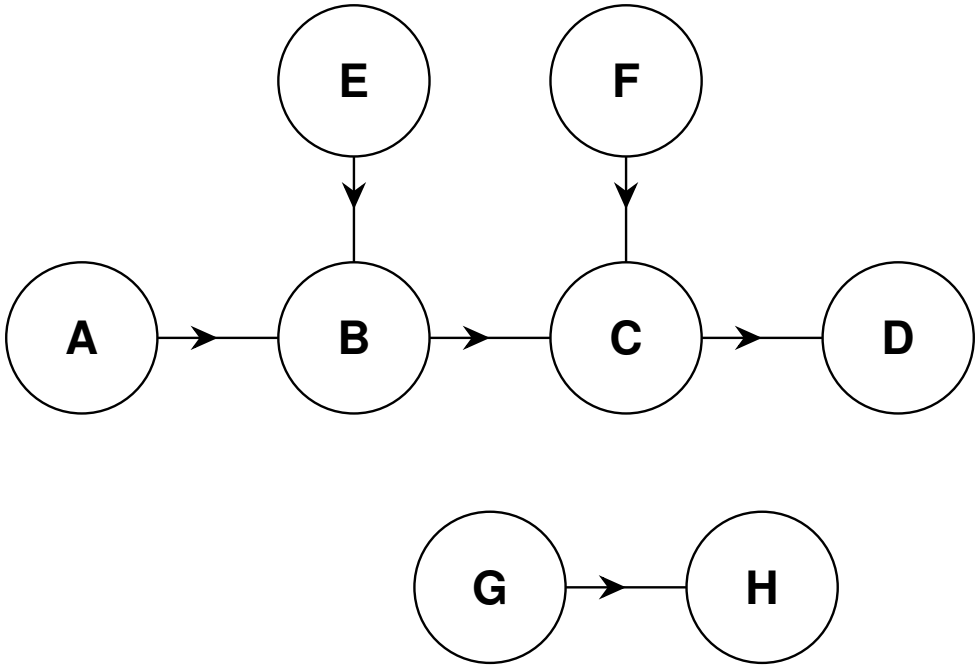
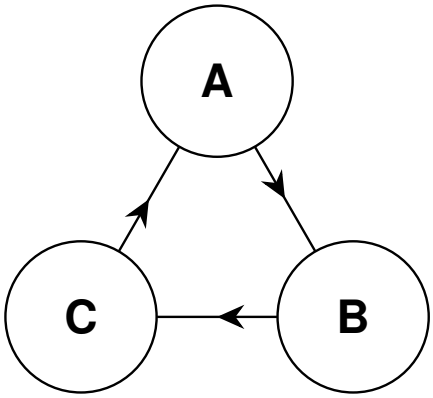
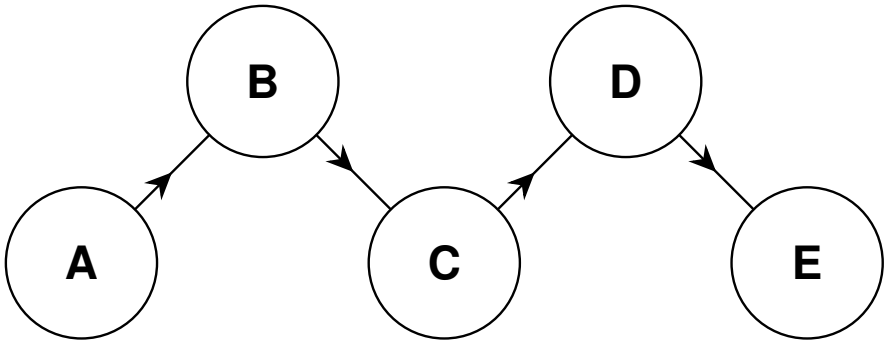
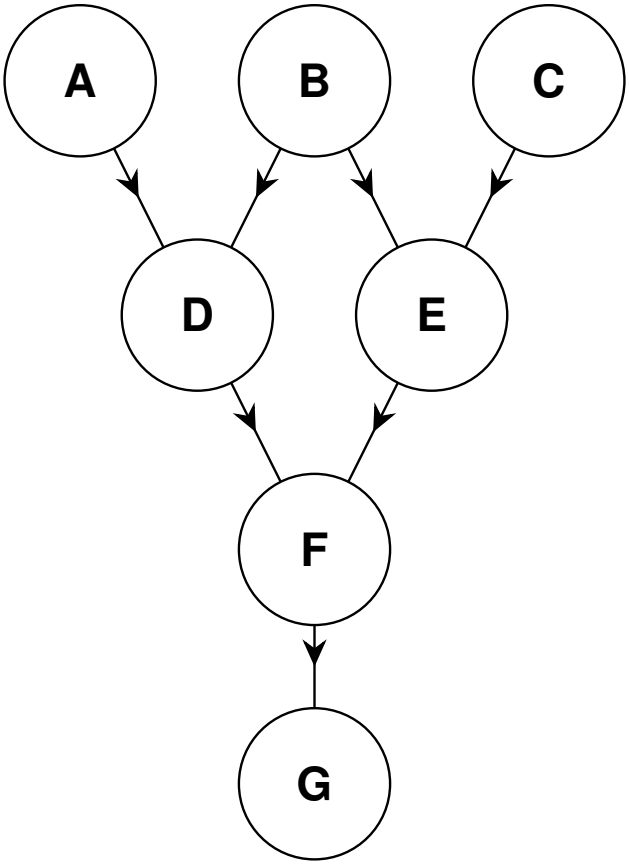
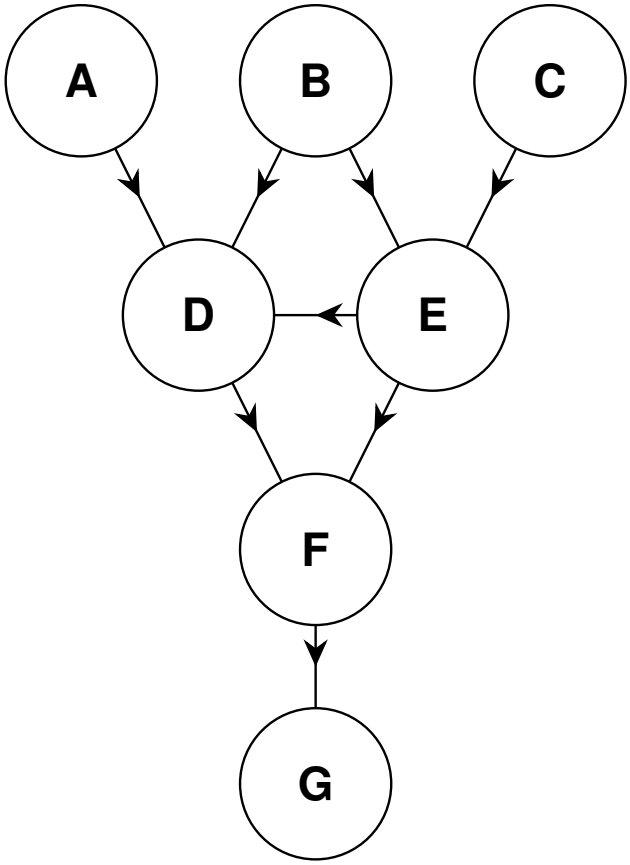



Lay the headers in a row, 1, 2, 3, 4, where 1 comes first. Put each card under a header, and a counter with that header's number on the card's circle. An arrow from one card to another means the first card must come earlier, under a smaller number. Cards with no arrow between them may share a header.




**Problem 9:** Put every card under a header, using as few headers as you can.



**Problem 10:** Can the cards of each network be put under the four headers?